Seaweeds Custom Dithering

Fullscreen Dithering and
dithered Materials
for Unity 6 URP

Contents

1 Introduction 1

2 Setup 1

2.1 Importing the package 1

2.2 Fullscreen Effect 2

2.2.1 Special case: Stacked Cameras 4

2.3 Materials 5

2.4 Textures 5

2.4.1 Threshold Maps 5

2.4.2 Color Palettes 6

2.4.3 Noise Textures 7

3 Material Settings 7

3.1.1 Fullscreen Material 7

3.1.2 Textured Dither 8

3.1.3 Colored Dither 9

3.1.4 Palette Dither 10

4 Finishing Up 10

4.1 Crediting and distribution 10

5 Credits 10

1 Introduction

Seaweed's Dithering is a collection of shaders to easily implement visual dithering for Unity URP projects. It includes a fullscreen render feature for screenspace dithering, to emulate the look of retro 3D games, as well as different material shaders to project dithering on the surfaces of 3D objects. The shaders provide many ways to tweak the look of the dithering so it can be customized for each project or desired style.

Note: This package was designed and tested for Unity 6 3D URP. If you want to use it for 2D or HDRP or another version of Unity, the package might not work or require major changes. However, if you have the experience or knowledge to make these changes, feel free to do so. I will note the places where URP or Unity 6 specific settings or features are used, which could cause problems in other versions, so you know what to watch out for.

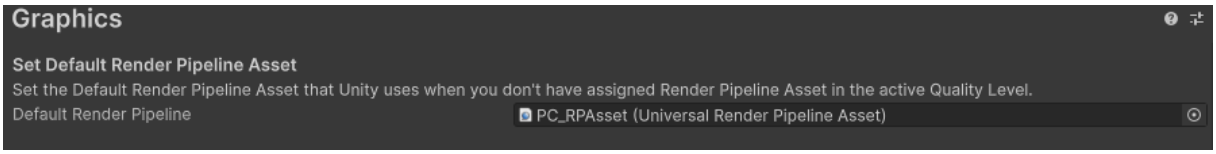
2 Setup

This setup guide will cover the implementation of the package on a stock URP project, but the package can be added at any point during development, so some steps that are covered in this guide might have been completed by you at some stage on your own. To be safe I recommend you follow the guide anyway to check if you have completed all the necessary steps. The setup guide assumes URP is already set up and running in your project, if it isn't you must setup URP in your project beforehand!

IMPORTANT! Some steps in this setup guide are mandatory for correct function of the package and if they are not setup in exactly the way shown the effect will not work as intended. If you decide to change settings or the way some things are done you may do so on your own discretion. If you have experience with shaders or the technical side of Unity, feel free to experiment or change things to your liking.

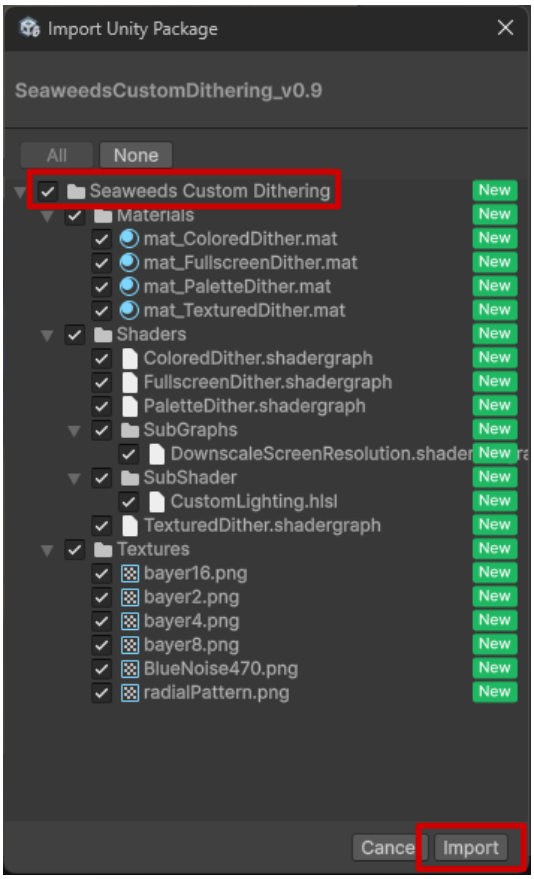
2.1 Importing the package

The setup of the fullscreen effect requires the most work, in fact it is the only part of the package that requires actual changes in the project or rendering settings. Before importing the package and starting the setup, check that the correct URP Asset is set in your Graphics settings under *Edit -> Project Settings -> Graphics "Default Render pipeline"*



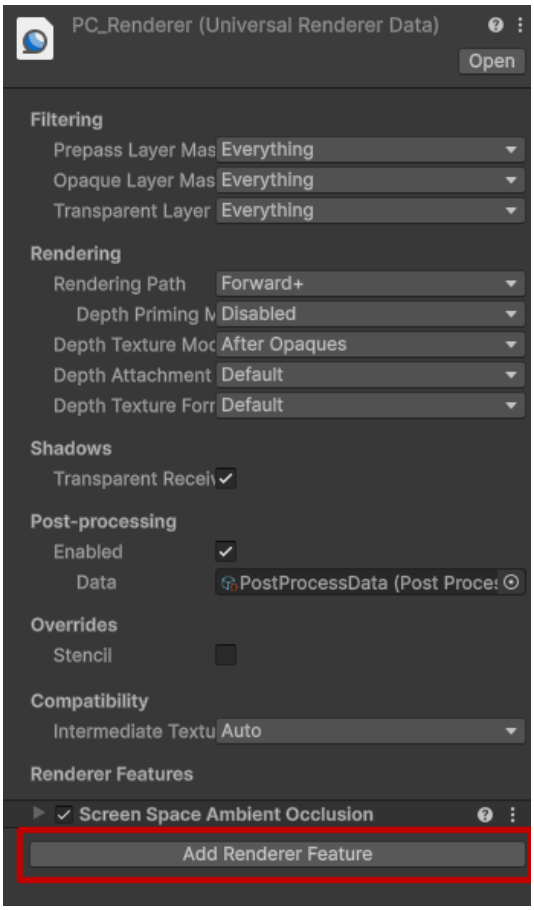
To import the package, simply drag and drop the file in your *Assets* Folder in the *Project* Window of your Unity Project. Simply click import and wait for Unity to finish the process. A new folder named *Seaweeds Custom Dithering* should appear in your *Assets*

folder. This where all the files for the package are stored, so when I mention something is included with the package it will be found here. The package comes with four example materials for every type of dithering. You can use these materials out of the box, tweak them, or create your own materials. Otherwise it contains the Shader Graphs, some example textures and settings for you to use or reference.

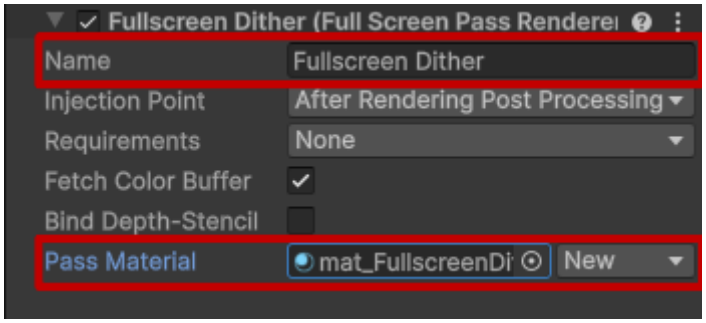


2.2 Fullscreen Effect

To use the fullscreen effect you need to make some changes to the renderer used by the URP Asset, which should receive the effect. By default, the URP Asset, as well as the renderers, are found in the *Settings* Folder and is called *PC_RPAsset*. If you don't know which URP Asset is actively used by Unity, you can look it up in the Project Settings, as mentioned earlier. In the Inspector view of the URP Asset, on top there is a list of active Renderers for this asset. Double-clicking a renderer opens the inspector for it. By default, the main URP renderer is called *PC_Renderer* and is located next to the URP Asset. Here you need to add a *new renderer feature* at the bottom and select *Full Screen Pass Renderer Feature.* You can reference or simply use the renderers included with the package located in the *Settings* folder.



Extend the new tab, change the name to Fullscreen Dither, or something that helps you identify it, and change the material to *mat_FullscreenDither* or your own material, which uses the *FullscreenDither* Shader. The effect should be working now and is ready to be used. If it doesn't check your material settings, which are explained further below.

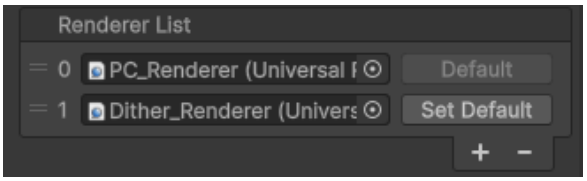


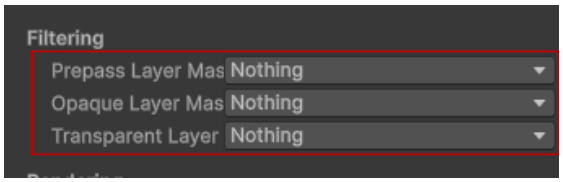
Note: If you don't see the effect, or for that matter any post-processing, make sure the "Post-Processing" Option on your camera is checked. Also, when using the fullscreen effect, it will be visible in scene view, which might be annoying for some. In this case you can simply disable the feature in the renderer but don't forget to re-enable it later.

2.2.1 Special case: Stacked Cameras

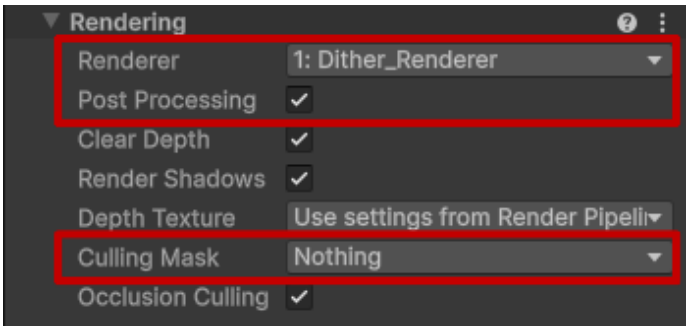
IMPORTANT! If you use stacked cameras to render different layers, you might get weird artifacts in the dither pattern. This is caused by the cameras using the same renderer, which causes the effect to be applied multiple times.

A workaround to this would be to create a new renderer solely for the fullscreen effect. (For clarification I will refer to this renderer as *renderer_n*). To do that either use the renderer that is included with the package or copy the currently used Renderer (*renderer_o*). If you already added the fullscreen effect to *renderer_o*, disable or delete it. On the *renderer_n*, disable or remove all effects, except the fullscreen dither. Set all Filtering options to *Nothing.*





Attach *renderer_n* to the list of renderers in the URP asset. Create a new camera and change its type from *Base* to *Overlay.* Set the *Culling Mask* to “Nothing”, and the Renderer to *renderer_n* and turn on *Post Processing*.



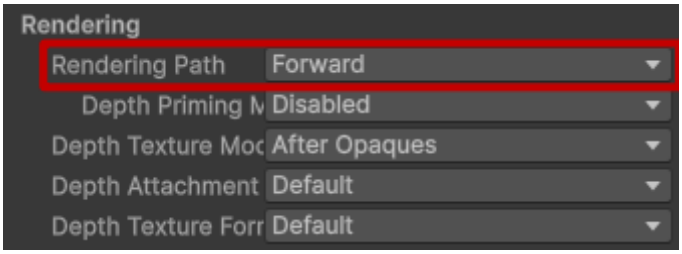
IMPORTANT! If you’re using multiple URP Assets, like for Quality settings, renderer_n needs to be attached to all relevant URP assets, or it won’t work with certain quality settings. You may notice this when setting up the new camera, that renderer_n won’t show up in the list of available renderers for the camera. You can check your current quality preset under Edit -> Project Settings -> Quality.

IMPORTANT! Using this workaround may result in a noticeable performance impact and a frame loss of up to 10 FPS, since an entire new camera is rendering in the scene!

Note: If you use this way, the dithering in scene view will disappear but remain visible in game view.

2.3 Materials

IMPORTANT! Before continuing with the materials an important note. These materials will only work correctly when the rendering path in the main renderer is set to "Forward" or "Deferred". The "+" versions of these paths is currently not working, as the way light is calculated in the shaders does not support them yet. If you are knowledgeable enough about HLSL or the way custom lighting is done in Unity, feel free to make the necessary changes to allow these rendering paths to be used.



IMPORTANT! The materials currently don't support metallic or specular, as well as smoothness. The resulting materials will only contain color and normal data; however, this produces a look closer to retro 3D games by which this package is inspired. Additionally, it is recommended you set the Filter Mode for your textures to point, to get sharp pixel edges. This will make the effect look better than on filtered textures.

The package comes with three different material shaders to allow for different looks of the dithering on objects. These shaders mostly work the same but are separated for different use cases so the materials don't get too bloated. Descriptions of the settings for these materials are covered in chapter 3.

2.4 Textures

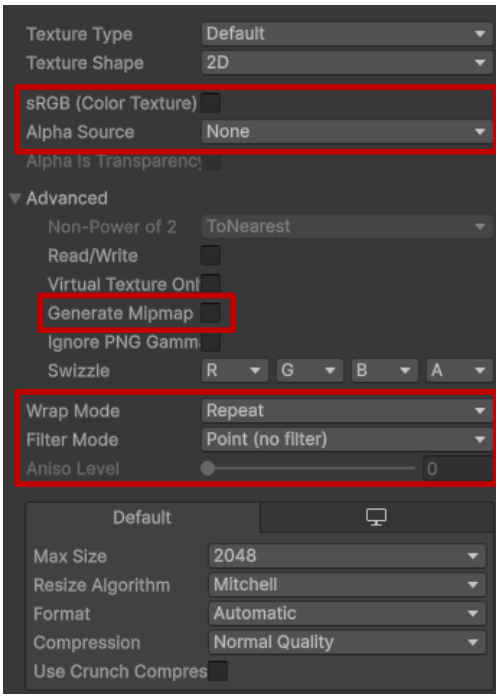
As mentioned, if you want to use your own textures for threshold maps or color palettes, these textures need to be set up in specific ways to ensure they are working correctly. To change the settings for textures, select the texture and change the specified values in the inspector window. Don't forget to apply the changes when you're finished! You can also reference the textures that come with the package to compare your settings.

Note: As mentioned, I recommend setting the Filter Mode of your base textures and normal maps to "Point (no filter)", otherwise the effect may look off.

2.4.1 Threshold Maps

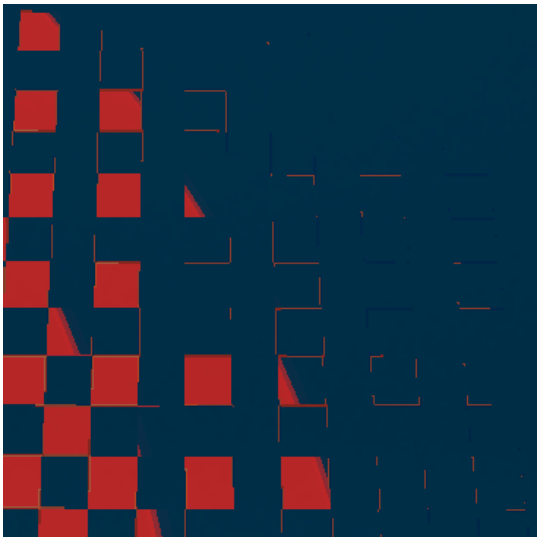
For threshold maps it is important to uncheck *sRGB* and set *Alpha Source* to "None". On the bottom set *AnisoLevel* to 0, then extend the *Advanced* tab and uncheck *Generate Mipmaps*. Lastly make sure *Wrap Mode* is set to "Repeat" and *Filter Mode* is set to "Point (no filter)". These settings ensure the threshold maps are interpreted and portrayed

correctly. If the displayed patterns look wrong or have artifacts, make sure these settings are applied.



2.4.2 Color Palettes

To make color palettes work correctly you absolutely must uncheck *Generate Mipmaps.* Failing to do this will cause some horrible artifacts between the pixels of the dither pattern, where a wrong mipmap is sampled.



Mipmaps enabled



Mipmaps disabled

Additionally, you should set *Wrap Mode* to “Clamp” to avoid repeating of the color palette and thus sampling of very bright values to dark colors. Also depending on what kind of color palette you want to use it might be sensible to set *Filter Mode* to “Point (no filter)”.

This setting is useful if you use a color palette with hard cutoffs between different colors. If using a gradient however, using “Bilinear” or “Trilinear” might be preferred.

2.4.3 Noise Textures

The settings of noise textures are not really important, but I recommend setting *Filter Mode* to “Point (no filter)”.

3 Material Settings

3.1.1 Fullscreen Material

Note: If you want your objects to look more pixelated, you need to reduce the max resolution in the BaseTexture’s settings.

ThresholdMap

> This slot contains the used threshold map for the dither effect and determines the look of the produced pattern. The package contains some Bayer matrices to create classic dither looks but any grayscale image can be used. These images need to be set up in a certain way, so the effect works correctly.

UseThresholdMap

> Toggles whether the effect uses the custom threshold map or the built-in pattern.

DitherSpread

> Controls how strongly the pattern is visible and how far it spreads.

ColorLevel

> Controls how many colors are available to portray the image. These values do not represent the total amount of visible colors in the image.
>
> *Note: Higher values will decrease the visibility of the dither pattern!*

Pixelation

> Reduces the resolution of the image in steps to emulate lower screen resolution. This effect is stable across all screen resolutions, so the pixels on Full-HD will be the same size as on UHD.

Brightness

> Lowers or increases brightness of the image, especially useful when using grayscale or color palettes.

Greyscale

Sets saturation to 0 transform image to grayscale.

ColorPalette

Takes in a texture to map image brightness to certain colors. Colors on the left side of the texture are used for dark values, colors to the right are used for bright values. Textures for color palette need to be set up in a specific way to work correctly.

UseColorPalette

Toggles whether color palette is used.

NoiseTexture

Takes in a greyscale image, preferably some sort of noise pattern, Blue Noise is recommended. The image is used additionally to threshold map to break up the dither pattern and add more variance. A Blue Noise Texture is included in the package.

UseRandomNoise

The effect comes with a built-in White Noise; this can be used instead of the Noise Texture.

UseNoise

Toggles whether noise should be used.

Note: *Leaving this checked, while the Noise Texture is empty and UseRandomNoise is unchecked, the pattern will not look correct. Make sure this option is only checked if either noise option is enabled.*

3.1.2 Textured Dither

Note: I will only go over settings that are different from the ones already covered previously.

ColorDepth

Same as *ColorLevel* in fullscreen material.

BaseColor

Can be used to recolor *BaseTexture*, leave white to disable.

BaseTexture

Main texture of the material, used for color.

NormalTexture

Slot for normal map.

NormalStrength

Controls intensity of normal map.

UseNormalMap

Toggles whether normal map should be used.

Note: If not using normal map, uncheck this option or set NormalStrength to 0. Otherwise, lighting will break on the material.

EmissionMap

Takes grayscale image to control where EmissionColor should be visible. If left empty, EmissionColor will affect the entire object.

EmissionColor

Controls color and intensity of emission. Leave black with 0 intensity to disable.

DitherTexture

Same as *ThresholdMap* in fullscreen material.

DitherSpread

Same as in fullscreen material.

DitherScale

Controls size of Dither pattern. Needs to be adjusted depending on size of *DitherTexture*.

AmbientStrength

Controls intensity of ambient lighting and brightness in places where no light hits the object.

ShadowCutoffSize

Controls the sharpness of shadow edges. 0 produces a perfectly sharp edge, higher values cause a larger area where shadow fades to light.

3.1.3 Colored Dither

Note: I will only go over settings that are different from the ones already covered previously.

BaseColor:

Controls main color of object.

DitherColor:

Controls color of dither pattern and fully dithered areas of the object.

3.1.4 Palette Dither

Note: I will only go over settings that are different from the ones already covered previously.

ColorPalette

Same as in fullscreen effect.

4 Finishing Up

I heavily encourage you to just play around with the different effects to get a feel for how the settings affect the look of the dithering and each other, as many of them will also affect each other. Should you feel the need to make changes to the shaders themselves you may do so freely, in fact I also want to encourage people who are smarter than me to improve on this base. In the end this package is supposed to offer you a base to easily integrate some basic dithering for your games and is meant to be customized to your personal needs.

4.1 Crediting and distribution

If you expand upon this package with new features, like HDRP support or something similar, and wish to distribute your version of this effect you may do so, but please credit the original work clearly and visibly. You may also include this effect in collections, like a retro effects package, in this case also simply credit the original work.

5 Credits

During the development of this package, I referenced many resources and tutorials by other creators, whom I want to thank for their work and inspiration. Especially I want to thank *Daniel Ilett,* whose tutorials helped me gain an understanding of shaders and Shader Graph, as well as writing HLSL and using it in combination with Shader Graph. Also, I want to thank *Gamelogic* and *Surma* for their great articles and videos about the workings of dithering and ways to implement it. And the countless people whose knowledge helped me along the way.

Daniel Ilett

Gamelogic

Surma

Threshold Maps